(1) Find the radian measure of the angle between hour-hand and minute-hand of a clock at :

(i) Twenty minutes past two (ii) Ten minutes past eleven

(iii) Thirty minutes past six (iv) Five minutes past one

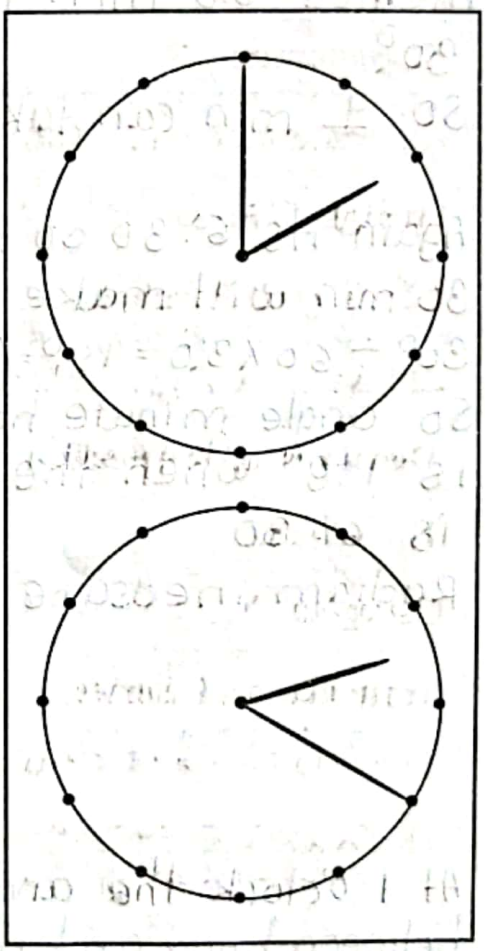
Solution : The speed of minute hand is 6° per minute. The speed of the hour hand is 0.5° per minute. The angle between hour hand and minute hand changes at the rate 5.5°.

(i) Twenty minutes past two : 2:20

We consider two positions of hands. First at 2:00 and second at 2:20.

At 2:00 angle between them is 60°. This angle gradually decreases, becomes zero and then increases. From 2:00 to 2:20 (i.e. in 20 minute time) the minute hand moves through 120° and hour hand moves through 20 × 0.5° = 10°.

The minute hand makes 120° - 60° = 60° angle with the position of hour hand at 2:00. But hour hand has moved through 10°. The required angle is 60° - 10° = 50°.



(ii) Ten minutes past eleven : 11:10

The angle between any two consecutive numbers of a clock is $\frac{360}{12} = 30°$. If the minute hand is at 2 and hour hand is at 11, angle between them is $(3 \times 30°) = 90°$.

But 10 past eleven i.e. at 11:10, the hour hand is moved 10 minutes towards 12. In 60 minutes, the hour hand covers.

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30° therefore in 10 minutes,
it covers $\left(\frac{30}{6}\right)^\circ = 5^\circ$
$\therefore$ the angle between the two
hands of the clock at ten past eleven
$= 90^\circ - 5^\circ = 85^\circ = \left(85 \times \frac{\pi}{180}\right)^c = \frac{17\pi^c}{36}$

(iii) Thirty minutes past six : 6 : 30

when the time is 6:30
The hour hand just little left to the
minute one. As per
method 60 min takes
30°
So 1 min can take $\frac{30^\circ}{60}$

Again its 6:30 so that
30 min will make
$30^\circ \div 60 \times 30 = 15^\circ = \left(\frac{15\pi}{180}\right)^c = \frac{\pi^c}{12}$
So angle minute hand
is "15°" when the time
is 6:30
Radian measure $= \frac{\pi^c}{12}$

(iv) Five minutes past one : 1 : 05

At 1 o'clock the angle
between hour and minute
hand $= 30^\circ$. In 5 minutes,
The angle made by hour hand $= 5/2$
The angle between the hour
and minute hand $= 30^\circ - 30^\circ + 5/2$
$= 2.5^\circ = \left(2.5 \times \frac{\pi}{180}\right)^c = \frac{\pi^c}{72}$

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(2) Find the exact time when the angle between hour-hand and the minute-hand of a clock will be : (i) 44° for the first time after 12 O'clock. (ii) 59° for the first time after 6 O'clock.

Solution : The angle between hour hand and minute hand changes at the rate 5.5°.

The angle between them will be 11° after 2 minutes.

(i) For 11° angle it takes 2 minute time. How much time will it take for 44° angle ? $\frac{2\times 44}{11}=8$

∴ At 12:08 the angle between hour-hand and the minute-hand of a clock will be 44° for the first time after 12 O'clock.

(ii) 59° for the first time after 6 O'clock.

Solution :

Initially i.e at 6 o'clock angle is 180°
To get angle 59°
180°-59° = 121° movement required
between hr hand and minute ~~time~~ hand.
For 11° angle it takes 2 minutes time.
How much time will it take for 121° angle?
$\frac{2\times 121}{11} = 2\times 11 = 22$ minute.

∴ At 6:22 the angle between hour-hand and the minute hand of a clock will be 59° For the first time after 6 o'clock.

(3) A wire of length 10 cm is bent to form an arc of a circle of radius 4 cm. Find the radian and degree measure of the angle subtended by the arc at the center of the circle. Also find the area of the corresponding sector.

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Solution : Given $r = 4$, arc length = length of the wire = $s = 10$

We know that $s = r \times \theta$

$\therefore$ $10 = 4 \times \theta$

$\therefore$ $\theta = \frac{5}{2}$ radian. $= \left(\frac{5}{2} \times \frac{180}{\pi}\right)^0 = \left(\frac{450}{\pi}\right)^0$

Now, the area of the sector $= A = \frac{1}{2} r^2 \theta = \frac{1}{2} r \times 10 = \frac{1}{2} \times 4 \times 10 = 20$ sq.cm.

(4) Arrange following angles in ascending order : π^c, π^0, 3^c, $100°$, $\frac{\pi^c}{2}$

Solution : To Canuert angles in degrees:

$\pi^c = \left(\pi \times \frac{180}{\pi}\right)^0 = 180^0$

$\pi^0 = 3^0 8' 29.73''$

$3^c = \left(3 \times \frac{180}{\pi}\right)^0 = \left(\frac{540}{\pi}\right)^0 = 171.88^0$

$100^0 = 100^0$

$\frac{\pi^c}{2} = \left(\frac{\pi}{2} \times \frac{180}{\pi}\right)^0 = 90^0$

From above Conversion:

$3^0 8' 29.73'' < 90^0 < 100^0 < 171.88^0 < 180^0$

Ascending order is

$\pi^0, \frac{\pi^c}{2}, 100^0, 3^c, \pi^c$

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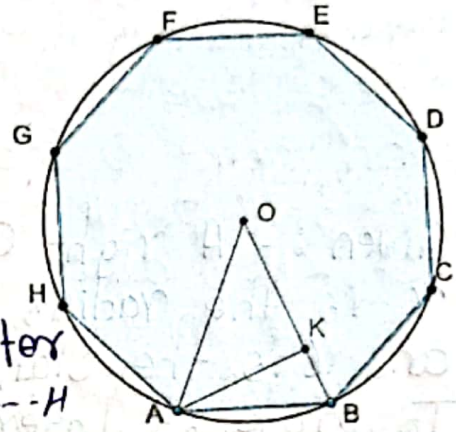
(5) ABCDEFGH is a regular octagon inscribed in a circle of radius 1 unit. O is the center of the circle. Find

(i) Radian measure of $\angle AOB$ (ii) $l(chord\ AB)$ (iii) $l(arc\ AB)$

(iv) Area of the region enclosed between chord AB and arc AB.

Solution : Draw AK perpendicular to OB.

Given OA = OB = 1 Find AK, OK, BK and then AB.



(i) Regular octagon has all its side equal if we join center O with all vertices A, B, C, ---- H

we will get OA = OB = Radius = 1

we will get 8 Segment

angle of each Segment $= \frac{360^0}{8} = 45^0$

$\therefore \angle AOB = 45^0$

ii) Draw $AK \perp OB$

In right angled ΔOKA

$\sin \angle ADK = \frac{AK}{OA}$

$\therefore \sin 45^0 = \frac{AK}{1}$

$\therefore AK = \frac{1}{\sqrt{2}}$ —— (i)

In ΔAOK

$\angle KOA = \angle KAO$

$\therefore AK = OK$

$\therefore OK = \frac{1}{\sqrt{2}}$

$AB = OK + BK$

$1 = \frac{1}{\sqrt{2}} + BK$

$\therefore BK = 1 - \frac{1}{\sqrt{2}}$ —— (ii)

In right angled ΔAKB by Pythagoras theorem

$AB^2 = AK^2 + BK^2$

$AB^2 = \left(\frac{1}{\sqrt{2}}\right)^2 + \left(1 - \frac{1}{\sqrt{2}}\right)^2$ (by (i) & (ii))

$\therefore AB^2 = \frac{1}{2} + 1 - \frac{2}{\sqrt{2}} + \frac{1}{2}$

$\therefore AB^2 = 2 - \frac{2}{\sqrt{2}} = 2 - \sqrt{2}$

$\therefore AB = \sqrt{2 - \sqrt{2}}$ unit

$\therefore AB = 0.765$ units.

$\therefore l(chord\ AB) = \sqrt{2-\sqrt{2}} = 0.765$ units

(iii) $l(arc\ AB) = r\theta$

$= 1 \times \frac{\pi}{4}$

$= \frac{\pi}{4}$ units

iv) Area of the region enclosed between chord AB and arc AB

= Arc area - ΔOAB area.

$= \frac{1}{2} r^2 \theta - \frac{1}{2} \times OB \times AK$.

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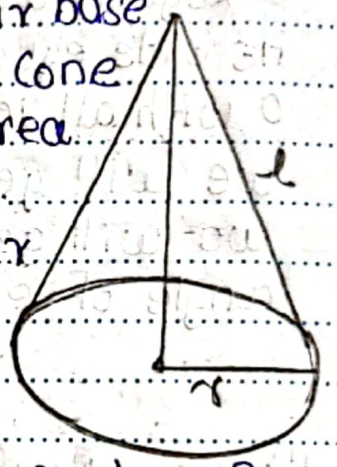
(6) Prove that the lateral surface area of a right circular cone is πrl, where r is the radius of the circular base and l is the slant height of the cone.

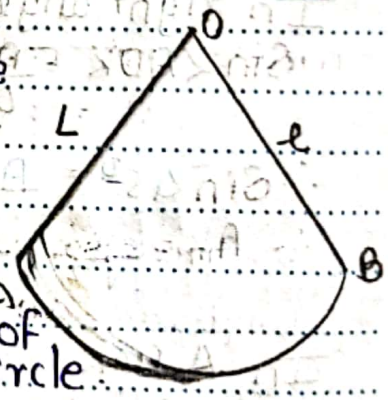
Solution :

Given :- A right circular cone with r is the radius of the circular base and l is the slant height of the cone.



To prove :- Lateral surface area or curved surface area $= \pi r l$

Activity :- If a perpendicular cut is made from a point on the circumference at the base to the vertex and the cone is opend out then the radius of a sector formed is l.



Since the circumference of the base $= 2\pi r$

lateral surface area of cone

$= \text{Area of Sector } OAB$

$= \dfrac{\text{Arc length of Sector}}{\text{circumference of circle}} \times \text{Area of circle}$

$= \dfrac{2\pi r}{2\pi r} \times \pi r^2$

$= \pi r l$

$\therefore$ Curved surface area or the lateral surface area of a right circular cone is $\pi r l$.

Sign of Teacher :

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